Algebra Tutorial

Subgroups, Normal Subgroups and Centers

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Problem 1. Determine which of the following subsets are subgroups of the given groups. Justify your answers.

(a) $2\mathbb{Z}$ and $3\mathbb{Z}$ in $(\mathbb{Z}, +)$;

(b) $\mathbb{Z}_{\geq 0}$ in $(\mathbb{Z}, +)$;

(c) $\mathbb{Q}^{\times}$ in $(\mathbb{R}^{\times}, \cdot)$;

(d) $\{z \in \mathbb{C} : |z| = 1\}$ in $(\mathbb{C}^{\times}, \cdot)$.

Problem 2. Let G be a group and let $H \leq G$. Define

$$C_G(H) = \{g \in G : gh = hg \text{ for every } h \in H\}.$$

Prove that $C_G(H)$ is a subgroup of G.

Problem 3. Let G be a group and let $a \in G$. Prove that

$$H = \{a^n : n \in \mathbb{Z}\}$$

is a subgroup of G. If G is finite, prove that H is finite.

Problem 4. Let

$$D_4 = \langle r, s \mid r^4 = s^2 = e,\ srs = r^{-1} \rangle$$

be the symmetry group of a square. Determine all subgroups of D_4 that have order 2.

Problem 5. Let G be a group and let $H \leq G$. Prove that H is normal in G if and only if

$$gHg^{-1} = H \qquad \text{for every } g \in G.$$

Use this criterion to determine whether

$$H = \{e, (12)\}$$

is normal in S_3.

Problem 6. Determine the center $Z(G)$ of each of the following groups:

(a) $(\mathbb{Z}, +)$;

(b) S_3;

(c) D_4, the symmetry group of a square;

(d) $GL_2(\mathbb{R})$, the group of invertible 2×2 real matrices.

Problem 7. Prove that the center

$$Z(G) = \{z \in G : zg = gz \text{ for every } g \in G\}$$

is a normal subgroup of G.

Problem 8. Let G be a group of order p^2, where p is prime. Prove that G is abelian.

Problem 9. Let G be a group and suppose that $G/Z(G)$ is cyclic. Prove that G is abelian.